MERN Setup, Development, and Deployment

These instructions are for Linux, specifically Ubuntu 22 was used.

Install Visual Studio Code
Install Chrome
Install Postman

--------- Install Node
sudo apt-get update
sudo apt-get install nodejs
sudo apt-get install npm

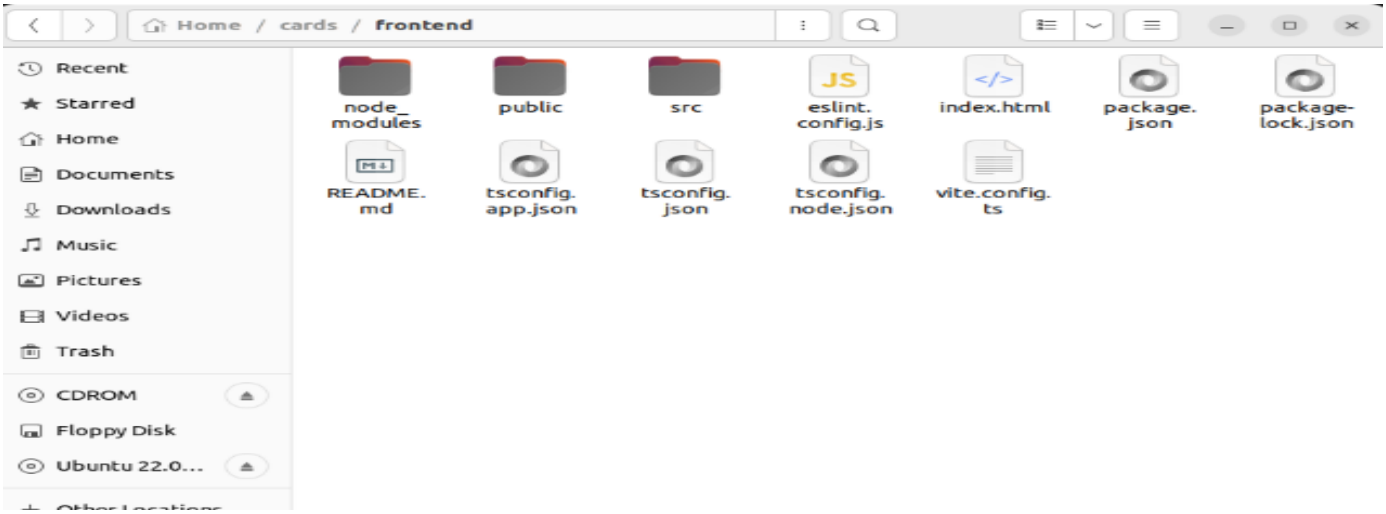
--------- update node
sudo npm cache clean -f
sudo npm install -g n
sudo n stable

------- install the react app creator and install nodemon (nodemon allows the server component to auto-restart after a change)
sudo npm install -g nodemon

Create app directory (suggest cards)
mkdir cards
cd cards

---------- create the **cards** app
In main directory (cards)

npm create vite@latest (It may ask you to install a package) – name it **frontend** when asked, select a **React** project, select **Typescript**
to test enter **cd frontend** and then **npm install** and then **npm run dev** then run a browser and load **http://localhost:5173**



Phase 1: A simple UI edit ----------

To frontend/src add **components** and **pages** directories

Open directory in Visual Studio Code. From cards directory type **code .** ← Yes the period/dot goes with the command.

Add two components

PageTitle.tsx (save into frontend/src/components)

```
function PageTitle()
{
   return(
     <h1 id="title">COP 4331 MERN Stack Demo</h1>
   );
};

export default PageTitle;
```

Login.tsx (save into frontend/src/components)

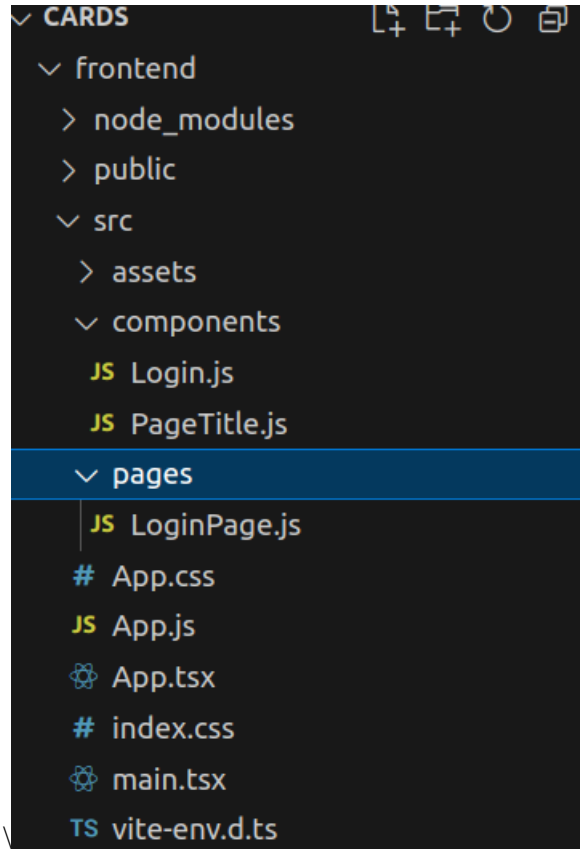
```
function Login()
{


  function doLogin(event:any) : void
  {
    event.preventDefault();

    alert('doIt()');
  }

    return(
      <div id="loginDiv">
        <span id="inner-title">PLEASE LOG IN</span><br />
        <input type="text" id="loginName" placeholder="Username" /><br />
        <input type="password" id="loginPassword" placeholder="Password" /><br />
        <input type="submit" id="loginButton" className="buttons" value = "Do It"
          onClick={doLogin} />
        <span id="loginResult"></span>
     </div>
    );
};

export default Login;
```

Add a page

LoginPage.tsx (save into pages)

```
import PageTitle from '../components/PageTitle.tsx';
import Login from '../components/Login.tsx';

const LoginPage = () =>
{

    return(
      <div>
        <PageTitle />
```

```
        <Login />
      </div>
    );
};

export default LoginPage;
```



Finally edit App.tsx

```
import './App.css';

import LoginPage from './pages/LoginPage.tsx';

function App()
{
  return (
    <LoginPage />
  );
}

export default App;
```

Test… (npm run dev) then run a browser and connect to http://localhost:5173/

Add backend –(phase1a)----------

Create server.js and package.json

Add the following to server.js (this goes into the root of the mern app, outside of the frontend directory)

```
const express = require('express');
const cors = require('cors');

const app = express();
app.use(cors());
app.use(express.json());

app.use((req, res, next) =>
{
  res.setHeader('Access-Control-Allow-Origin', '*');
  res.setHeader(
    'Access-Control-Allow-Headers',
    'Origin, X-Requested-With, Content-Type, Accept, Authorization'
  );
  res.setHeader(
    'Access-Control-Allow-Methods',
    'GET, POST, PATCH, DELETE, OPTIONS'
  );
  next();
});

app.listen(5000); // start Node + Express server on port 5000
```

Add the following to package.json

```
{
  "name": "backend",
  "version": "1.0.0",
  "description": "",
  "main": "server.js",
  "scripts": {
    "test": "echo \"Error: no test specified\" && exit 1",
    "start": "nodemon server.js"
  },
  "author": "",
  "license": "ISC",
  "dependencies": {
        "cors": "^2.8.5",
        "express": "^4.18.2",
        "mongodb": "^5.8.1"
  }
}
```

In root of the mern app directory
sudo npm install
sudo npm install express --save
sudo npm install mongodb
sudo npm install cors

(Might have to run **echo fs.inotify.max_user_watches=524288 | sudo tee -a /etc/sysctl.conf && sudo sysctl -p**)

Test… (npm start)

```
leinecker@ubuntu:~/cards$ npm start

> backend@1.0.0 start /home/leinecker/cards
> nodemon server.js

[nodemon] 2.0.4
[nodemon] to restart at any time, enter `rs`
[nodemon] watching path(s): *.*
[nodemon] watching extensions: js,mjs,json
[nodemon] starting `node server.js`
```

Phase 2: Add baseball card page to front end----------

Add component LoggedInName.tsx

```
function LoggedInName()
{

    function doLogout(event:any) : void
    {
          event.preventDefault();

        alert('doLogout');
    };

    return(
      <div id="loggedInDiv">
        <span id="userName">Logged In As John Doe </span><br />
        <button type="button" id="logoutButton" className="buttons"
           onClick={doLogout}> Log Out </button>
      </div>
    );
};

export default LoggedInName;
```

Add component CardUI.tsx

```
function CardUI()
{

    function addCard(event:any) : void
    {
          event.preventDefault();

          alert('addCard()');

    };

    function searchCard(event:any) : void
    {
        event.preventDefault();

         alert('searchCard');
    };
```

```
    return(
      <div id="accessUIDiv">
       <br />
       <input type="text" id="searchText" placeholder="Card To Search For" />
       <button type="button" id="searchCardButton" className="buttons"
           onClick={searchCard}> Search Card </button><br />
       <span id="cardSearchResult"></span>
       <p id="cardList"></p><br /><br />
       <input type="text" id="cardText" placeholder="Card To Add" />
       <button type="button" id="addCardButton" className="buttons"
          onClick={addCard}> Add Card </button><br />
       <span id="cardAddResult"></span>
     </div>
    );
}

export default CardUI;
```

Create page CardPage.tsx

```
import PageTitle from '../components/PageTitle';
import LoggedInName from '../components/LoggedInName';
import CardUI from '../components/CardUI';

const CardPage = () =>
{
    return(
        <div>
            <PageTitle />
            <LoggedInName />
            <CardUI />
        </div>
    );
}

export default CardPage;
```

Edit App.tsx

```
import './App.css';

import LoginPage from './pages/LoginPage.tsx';
import CardPage from './pages/CardPage.tsx';

function App()
{
  return (
    <CardPage />
  );
}

export default App;
```

Test… (npm run dev)

COP 4331 MERN Stack Demo

Logged In As John Doe

Log Out

Card To Search For | Search Card

Card To Add | Add Card

Routing (phase 2a)

-- from frontend folder
npm install react-router-dom

Edit App.tsx

```
import { BrowserRouter, Routes, Route } from "react-router-dom";
import './App.css';

import LoginPage from './pages/LoginPage';
import CardPage from './pages/CardPage';

function App() {
  return (
    <BrowserRouter>
      <Routes>
        <Route path="/" element={<LoginPage />} />
        <Route path="/cards" element={<CardPage />} />
      </Routes>
    </BrowserRouter>
      );
}

export default App;
```

Test… (npm run dev)

Input field persistence (phase 3)

(Might have to run **echo fs.inotify.max_user_watches=524288 | sudo tee -a /etc/sysctl.conf && sudo sysctl -p**)

To **Login.tsx** and **CardUI.tsx** add **import React, { useState } from 'react';**

Add to Login function

```
  const [message,setMessage] = useState('');
  const [loginName,setLoginName] = React.useState('');
  const [loginPassword,setPassword] = React.useState('');
```

Edit input fields for input data

```
Login: <input type="text" id="loginName" placeholder="Username"
          onChange={handleSetLoginName} />
Password: <input type="password" id="loginPassword" placeholder="Password"
          onChange={handleSetPassword} />
```

Edit span for output label

```
<span id="loginResult">{message}</span>
```

Add Two Functions

```
    function handleSetLoginName( e: any ) : void

    {

      setLoginName( e.target.value );

    }


    function handleSetPassword( e: any ) : void

    {

      setPassword( e.target.value );

    }
```

Edit doLogin

```
    function doLogin(event:any) : void
    {
        event.preventDefault();

        alert alert('doIt() ' + loginName + ' ' + loginPassword );
    };
```

Add to CardUI.tsx

```
import React, { useState } from 'react';
```

Add into CardUI function

```
    const [message,setMessage] = useState('');
    const [searchResults,setResults] = useState('');
    const [cardList,setCardList] = useState('');
    const [search,setSearchValue] = React.useState('');
    const [card,setCardNameValue] = React.useState('');
```

Edit UI as follows:

```
<div id="cardUIDiv">
  <br />
  Search: <input type="text" id="searchText" placeholder="Card To Search For"
    onChange={handleSearchTextChange} />
  <button type="button" id="searchCardButton" className="buttons"
    onClick={searchCard}> Search Card</button><br />
  <span id="cardSearchResult">{searchResults}</span>
  <p id="cardList">{cardList}</p><br /><br />
  Add: <input type="text" id="cardText" placeholder="Card To Add"
    onChange={handleCardTextChange} />
  <button type="button" id="addCardButton" className="buttons"
    onClick={addCard}> Add Card </button><br />
  <span id="cardAddResult">{message}</span>
</div>
```

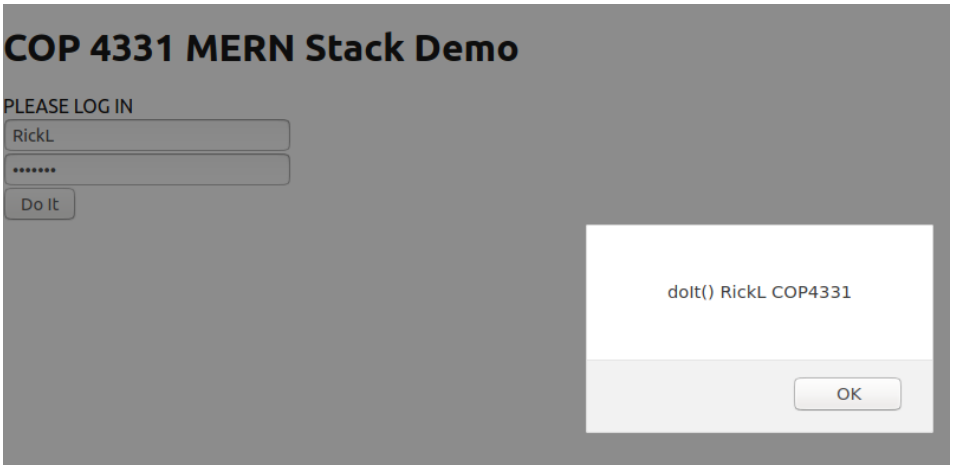
Add Two Functions

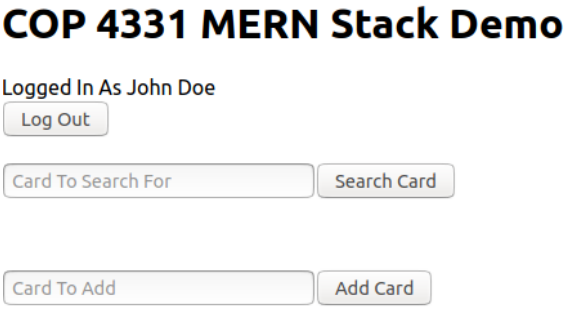
```
    function handleSearchTextChange( e: any ) : void
    {
        setSearchValue( e.target.value );
    }

    function handleCardTextChange( e: any ) : void
    {
        setCardNameValue( e.target.value );
    }
```

Edit addCard and searchCard as follows:

```
    function addCard(e:any) : void
    {
         e.preventDefault();

         alert('addCard() ' + card);

    };

    function searchCard(e:any) : void
    {
        e.preventDefault();

        alert('searchCard() ' + search);
    };
```

Test... (npm run dev)





API without DB access (phase 3a)

Open server.js

Add the following:

```
var cardList =
[
  'Roy Campanella',
  'Paul Molitor',
  'Tony Gwynn',
  'Dennis Eckersley',
  'Reggie Jackson',
  'Gaylord Perry',
  'Buck Leonard',
  'Rollie Fingers',
  'Charlie Gehringer',
  'Wade Boggs',
  'Carl Hubbell',
  'Dave Winfield',
  'Jackie Robinson',
  'Ken Griffey, Jr.',
  'Al Simmons',
  'Chuck Klein',
  'Mel Ott',
  'Mark McGwire',
  'Nolan Ryan',
  'Ralph Kiner',
  'Yogi Berra',
```

'Goose Goslin',
'Greg Maddux',
'Frankie Frisch',
'Ernie Banks',
'Ozzie Smith',
'Hank Greenberg',
'Kirby Puckett',
'Bob Feller',
'Dizzy Dean',
'Joe Jackson',
'Sam Crawford',
'Barry Bonds',
'Duke Snider',
'George Sisler',
'Ed Walsh',
'Tom Seaver',
'Willie Stargell',
'Bob Gibson',
'Brooks Robinson',
'Steve Carlton',
'Joe Medwick',
'Nap Lajoie',
'Cal Ripken, Jr.',
'Mike Schmidt',
'Eddie Murray',
'Tris Speaker',
'Al Kaline',
'Sandy Koufax',
'Willie Keeler',
'Pete Rose',
'Robin Roberts',
'Eddie Collins',
'Lefty Gomez',
'Lefty Grove',
'Carl Yastrzemski',
'Frank Robinson',
'Juan Marichal',
'Warren Spahn',
'Pie Traynor',
'Roberto Clemente',
'Harmon Killebrew',
'Satchel Paige',
'Eddie Plank',
'Josh Gibson',
'Oscar Charleston',
'Mickey Mantle',
'Cool Papa Bell',
'Johnny Bench',
'Mickey Cochrane',
'Jimmie Foxx',
'Jim Palmer',
'Cy Young',
'Eddie Mathews',
'Honus Wagner',
'Paul Waner',
'Grover Alexander',
'Rod Carew',

```
  'Joe DiMaggio',
  'Joe Morgan',
  'Stan Musial',
  'Bill Terry',
  'Rogers Hornsby',
  'Lou Brock',
  'Ted Williams',
  'Bill Dickey',
  'Christy Mathewson',
  'Willie McCovey',
  'Lou Gehrig',
  'George Brett',
  'Hank Aaron',
  'Harry Heilmann',
  'Walter Johnson',
  'Roger Clemens',
  'Ty Cobb',
  'Whitey Ford',
  'Willie Mays',
  'Rickey Henderson',
  'Babe Ruth'
];
```

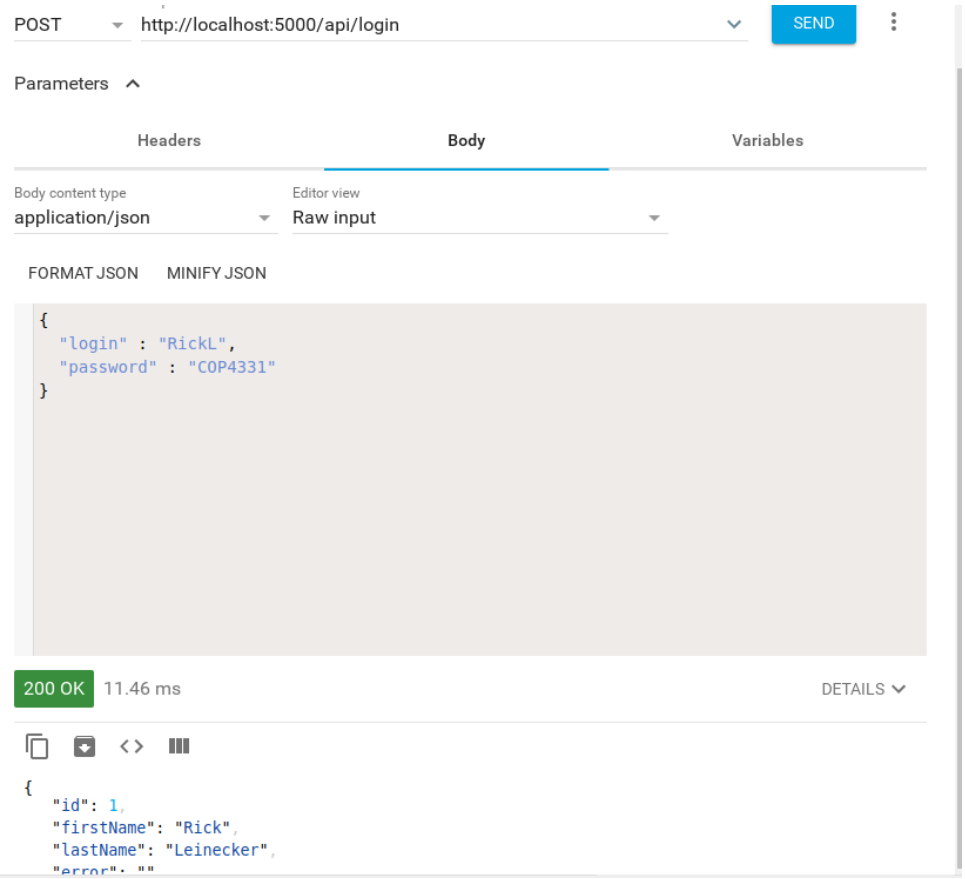
Now add three API endpoints

```
app.post('/api/addcard', async (req, res, next) =>
{
  // incoming: userId, color
  // outgoing: error

  var error = '';

  const { userId, card } = req.body;

  // TEMP FOR LOCAL TESTING.
  cardList.push( card );

  var ret = { error: error };
  res.status(200).json(ret);
});

app.post('/api/login', async (req, res, next) =>
{
  // incoming: login, password
  // outgoing: id, firstName, lastName, error

  var error = '';

  const { login, password } = req.body;

  var id = -1;
  var fn = '';
  var ln = '';

  if( login.toLowerCase() == 'rickl' && password == 'COP4331' )
```

```
  {
    id = 1;
    fn = 'Rick';
    ln = 'Leinecker';
  }
  else
  {
    error = 'Invalid user name/password';
  }

  var ret = { id:id, firstName:fn, lastName:ln, error:error};
  res.status(200).json(ret);
});

app.post('/api/searchcards', async (req, res, next) =>
{
  // incoming: userId, search
  // outgoing: results[], error

  var error = '';

  const { userId, search } = req.body;
  var _search = search.toLowerCase().trim();
  var _ret = [];

  for( var i=0; i<cardList.length; i++ )
  {
    var lowerFromList = cardList[i].toLocaleLowerCase();
    if( lowerFromList.indexOf( _search ) >= 0 )
    {
      _ret.push( cardList[i] );
    }
  }

  var ret = {results:_ret, error:''};
  res.status(200).json(ret);
});
```

Test with ARC or Postman

Enter the URL
Set to application/json content type
Set to POST
Click on body and then raw
Add JSON to the body



Remember to make doLogin async

Edit doLogin:

```
    async function doLogin(event:any) : Promise<void>
    {
        event.preventDefault();

        var obj = {login:loginName,password:loginPassword};
        var js = JSON.stringify(obj);

        try
        {
            const response = await fetch('http://localhost:5000/api/login',
                {method:'POST',body:js,headers:{'Content-Type':
'application/json'}});

            var res = JSON.parse(await response.text());

            if( res.id <= 0 )
            {
                setMessage('User/Password combination incorrect');
            }
            else
            {
                var user =
{firstName:res.firstName,lastName:res.lastName,id:res.id}
                localStorage.setItem('user_data', JSON.stringify(user));

                setMessage('');
```

```
                window.location.href = '/cards';
            }
        }
        catch(error:any)
        {
            alert(error.toString());
            return;
        }
      };
```

Test ...

Now for CardUI.tsx

```
    let _ud : any = localStorage.getItem('user_data');
    let ud = JSON.parse( _ud );
    let userId : string = ud.id;
    const [message,setMessage] = useState('');
    const [searchResults,setResults] = useState('');
    const [cardList,setCardList] = useState('');
    const [search,setSearchValue] = React.useState('');
    const [card,setCardNameValue] = React.useState('');

    async function addCard(e:any) : Promise<void>
    {
        e.preventDefault();

        let obj = {userId:userId,card:card};
        let js = JSON.stringify(obj);

        try
        {
            const response = await
fetch('http://localhost:5000/api/addcard',
            {method:'POST',body:js,headers:{'Content-Type':
'application/json'}});

            let txt = await response.text();
            let res = JSON.parse(txt);

            if( res.error.length > 0 )
            {
                setMessage( "API Error:" + res.error );
            }
            else
            {
                setMessage('Card has been added');
            }
        }
        catch(error:any)
        {
            setMessage(error.toString());
        }
```

```
    };

    async function searchCard(e:any) : Promise<void>
    {
        e.preventDefault();

        let obj = {userId:userId,search:search};
        let js = JSON.stringify(obj);

        try
        {
            const response = await
fetch('http://localhost:5000/api/searchcards',
            {method:'POST',body:js,headers:{'Content-Type':
'application/json'}});

            let txt = await response.text();
            let res = JSON.parse(txt);
            let _results = res.results;
            let resultText = '';
            for( let i=0; i<_results.length; i++ )
            {
                resultText += _results[i];
                if( i < _results.length - 1 )
                {
                    resultText += ', ';
                }
            }
            setResults('Card(s) have been retrieved');
            setCardList(resultText);
        }
        catch(error:any)
        {
            alert(error.toString());
            setResults(error.toString());
        }
    };
```

Test...

COP 4331 MERN Stack Demo

Logged In As John Doe
Log Out

ba | Search Card
Card(s) have been retrieved

Ernie Banks, Barry Bonds, Babe Ruth

Card To Add | Add Card

Create MongoDB DB

Go to website and show

Make sure to go to network security and open for all IPs

Add to server.js (PHASE 4)

```
const MongoClient = require('mongodb').MongoClient;
const url = 'mongodb+srv://RickLeinecker:COP4331Rocks@cluster0-
4pisv.mongodb.net/COP4331?retryWrites=true&w=majority';

const client = new MongoClient(url);
client.connect();
```

Change api/login

```
app.post('/api/login', async (req, res, next) =>
{
  // incoming: login, password
  // outgoing: id, firstName, lastName, error

 var error = '';

  const { login, password } = req.body;

  const db = client.db('COP4331Cards');
  const results = await
db.collection('Users').find({Login:login,Password:password}).toArray();

  var id = -1;
  var fn = '';
  var ln = '';

  if( results.length > 0 )
  {
    id = results[0].UserID;
    fn = results[0].FirstName;
    ln = results[0].LastName;
  }

  var ret = { id:id, firstName:fn, lastName:ln, error:''};
  res.status(200).json(ret);
});
```

Edit LoggedInName

```
function LoggedInName()
{

    function doLogout(event:any) : void
    {
      event.preventDefault();

      localStorage.removeItem("user_data")
      window.location.href = '/';
```

```
  };

    return(
      <div id="loggedInDiv">
        <span id="userName">Logged In As John Doe </span><br />
        <button type="button" id="logoutButton" className="buttons"
           onClick={doLogout}> Log Out </button>
      </div>
    );
};
```

Edit api/addcard

```
app.post('/api/addcard', async (req, res, next) =>
{
  // incoming: userId, color
  // outgoing: error

  const { userId, card } = req.body;

  const newCard = {Card:card,UserId:userId};
  var error = '';

  try
  {
    const db = client.db('COP4331Cards');
    const result = db.collection('Cards').insertOne(newCard);
  }
  catch(e)
  {
    error = e.toString();
  }

  cardList.push( card );

  var ret = { error: error };
  res.status(200).json(ret);
});
```

Edit api/searchcard

```
app.post('/api/searchcards', async (req, res, next) =>
{
  // incoming: userId, search
  // outgoing: results[], error

  var error = '';

  const { userId, search } = req.body;

  var _search = search.trim();

  const db = client.db('COP4331Cards');
  const results = await db.collection('Cards').find({"Card":{$regex:_search+'.*', $options:'i'}}).toArray();
```

```
  var _ret = [];
  for( var i=0; i<results.length; i++ )
  {
    _ret.push( results[i].Card );
  }

  var ret = {results:_ret, error:error};
  res.status(200).json(ret);
});
```